

# Course-End Project Writeup

## Creating Designs by Leveraging OpenAI and Gradio UI

**Program:** IITK AIML – Advanced Generative AI (Course 6)

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### 1. Situation

Creative teams need fast, high-quality visual concepts for Netflix-style digital marketing banners and posters. Traditional design workflows are slower for early campaign ideation and A/B creative exploration.

### 2. Task

Build a web platform that converts natural-language prompts into bespoke marketing designs using OpenAI image generation and a Gradio UI, so marketers can prototype campaign visuals in seconds.

### 3. Action

- **Libraries:** Used `openai`, `gradio`, `PIL`, and related helpers.
- **`generate_image` function:** Accepts a text prompt and size, calls OpenAI Images API (`gpt-image-1` — OpenAI image generation available on this account), decodes the returned image, and returns a PIL image.
- **Gradio UI:** Built `gr` . Interface with prompt input, size selector, and generated image output.
- **Campaign examples:** Generated sample Netflix-style posters (thriller / neon city “DARK”, romantic drama series in rain) to validate the end-to-end flow.

### 4. Result

Designers can prototype campaign visuals from prompts in seconds. The Gradio app supports iterative creative exploration and reduces time-to-concept for promotional assets. Sample outputs demonstrate thriller and romance campaign styles suitable for streaming marketing.

### 5. Tools & stack

Python · Jupyter Notebook · OpenAI Images API (`gpt-image-1`) · Gradio · Pillow

### 6. Deliverables uploaded

- Writeup (this PDF)
- Screenshots of Gradio UI and generated Netflix-style designs

- Source code notebook (`netflix_design_generator.ipynb`)

*Note: Assignment materials reference DALL·E; this submission uses the OpenAI Images API model available on the account (gpt-image-1) with the same Gradio workflow.*